

Solution- Exercice 02 (part 1, 2 and 3)

Exercice 02:

A driving school, established in Algiers for several years, has not stopped developing on the national territory. Therefore, its leaders decided to create agencies in the cities of Constantine and Oran.

The centralized database that was initially developed has the following structure:

AGENCY (NumAG, cityAG, Num-manager*, Nbr-instructors)

ADMINISTRATIVE_STAFF (NumS, AddressS, NameS, NumAG*, Date of birth, Function)

VEHICLE (NumV, Brand, Type, NumAG*, Num-Instructor*)

CUSTOMER (NumC, NameC, AddressC, NumAG*, Age)

COURSE (Num-Instructor*, NumC*, Course-Date, NumV*)

EXAM (NumC*, ExamDate, Result)

Remarques

- The manager and instructors are part of the administrative staff.
- A vehicle is assigned to an instructor
- The school wants to create three sites to host data relating to the agencies of each city: Algiers, Oran and Constantine Sites

1. Fragmentation

- Propose a fragmentation scheme for this database taking into account hypotheses related to sites. Give the algebraic expressions for each table.

Table	Fragment/ Algebraic Expression
Agency « HPF »	Agency_Alger= $\sigma_{Ville=Alger} (Agency)$ Agency_Oran= $\sigma_{Ville=Oran} (Agency)$ Agency_Constantine= $\sigma_{Ville=Canstantine} (Agency)$
Administratif-Staff « HDF »	Administratif-Staff_Alger= Administratif-Staff $\bowtie$ Agency_Alger Administratif-Staff _Oran= Administratif-Staff $\bowtie$ Agency_Oran Administratif-Staff_Constantine= Administratif-Staff $\bowtie$ Agency_Constantine

Vehicle «HDF »	Vehicle_Alger= Vehicle ⋈ Agency_Alger Vehicle _Oran= Vehicle⋈ Agency_Oran Vehicle _Constantine= Vehicle ⋈ Agency_Constantine
Customer «HDF»	Client_Alger= Customer ⋈ Agency_Alger Client _Oran= Customer ⋈ Agency_Oran Client _Constantine= Customer ⋈ Agency_Constantine
Exam «HDF»	Exam_Alger= Exam ⋈ Client_Alger Exam _Oran= Exam ⋈ Client _Oran Exam _Constantine= Exam ⋈ Client _Constantine
COURS «HDF»	Cours_Alger= Cours ⋈ Client _Alger Cours _Oran= Cours ⋈ Client _Oran Cours _Constantine= Cours⋈ Client _Constantine

b) What are the fragmentation possibilities of the COURSE table. Give the best solution.

Solution 1: DHF of the table ADMINISTRATIF-Staff

Solution 2 : DHF of la table CUSTOMER

Solution 3: DHF of la table VEHICLE

Best solution : The three solutions lead to the same result because the tables VEHICLE, CUSTOMER AND ADMINISTRATIVE staff are fragmented on the same foreign key (NumAG).

c) Give the SQL queries corresponding to the fragmentation of the relations AGENCY and Customer.

```
Create table AGENCY (NumAG integer , CityAG varchar(50),  
NumManager integer, Nbr-instructorrs integer, constraint pk_agence  
primary key (NumAG ))
```

```
Partition by List (CityAG)(
```

```
Partition AGENCY_Alger VALUES ('Alger'),
```

```
Partition AGENCY_Oran VALUES ('Oran'),
```

```
Partition AGENCY_Constantine VALUES ('Constantine')
```

```
);
```

```
Create table CUSTOMER (NumC integer , NameC varchar(50) ,  
AdresseC varchar(100), NumAG integer, Age integer, constraint  
pk_client primary key (NumC ) ,  
  
constraint fk_agence foreign key(NumAG) refernces agence(NumAG))  
  
Partition by reference (fk_Agence) ;
```

d. Oracle fragments a table when it is created. Give a solution to fragment a table already containing data. Apply the solution to the AGENCE table

1- Create a new table **Agency_Frag** as done in previous script (with partitionning) .

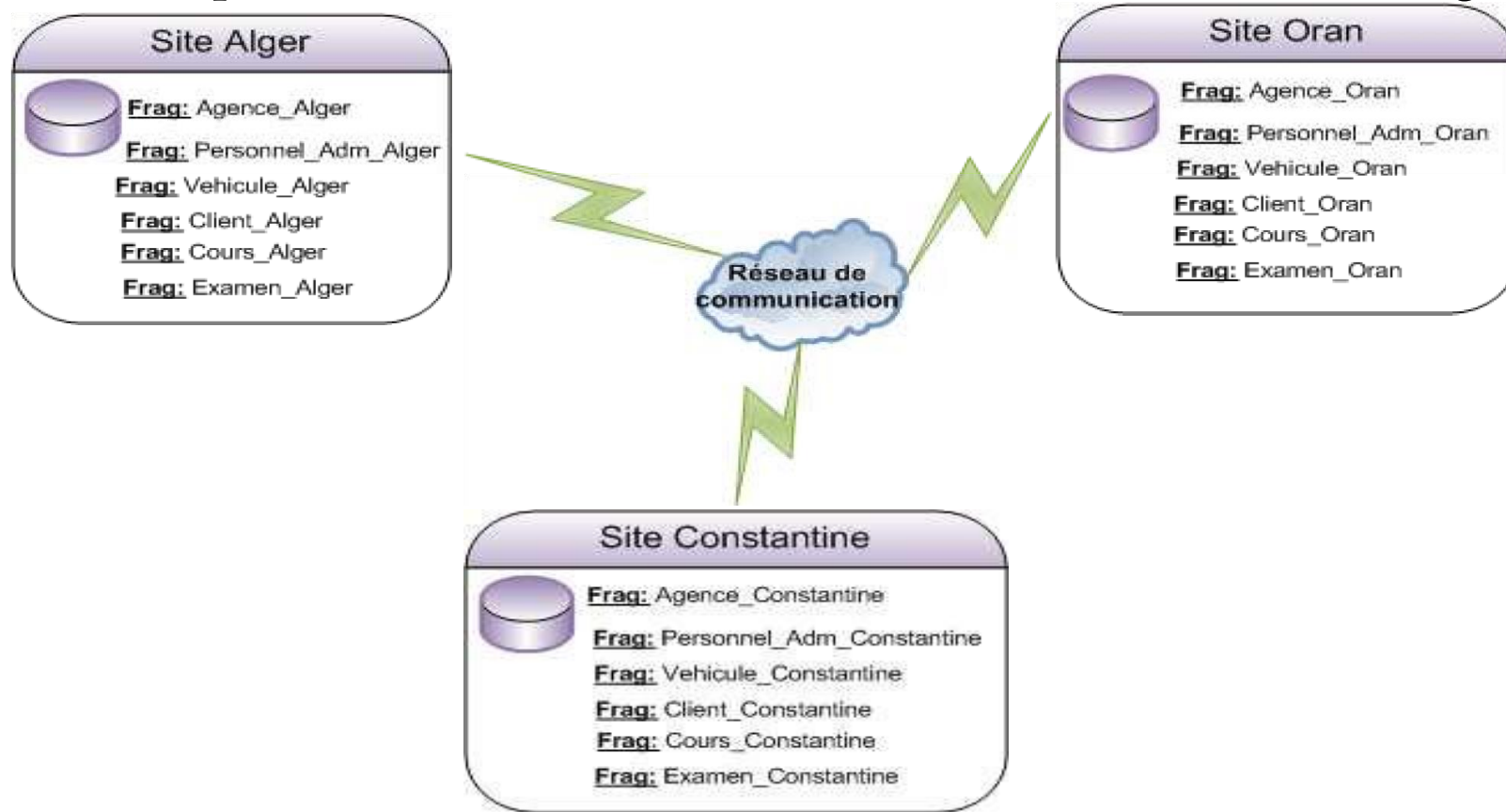
2- Load data : Insert into Agency_Frag select * from Agency

3- Drop Table Agency

4- Rename the fragmented table : Alter Table Agency_Frag rename to Agence

2. Allocation

a. Propose an allocation schema for all fragments.



b. Vehicles belonging to agencies in the wilaya of Algiers are very often accessed from Oran.

- Propose an improvement to the allocation scheme by taking this constraint into account.

- Replicate (create a copy): of the fragment Vehicle_Alger.

- Give the disadvantages of this solution.

Update problems: Each time a change is made on Oran, Algiers or Oran site, the other fragment must be updated in real time.

C. Evaluation

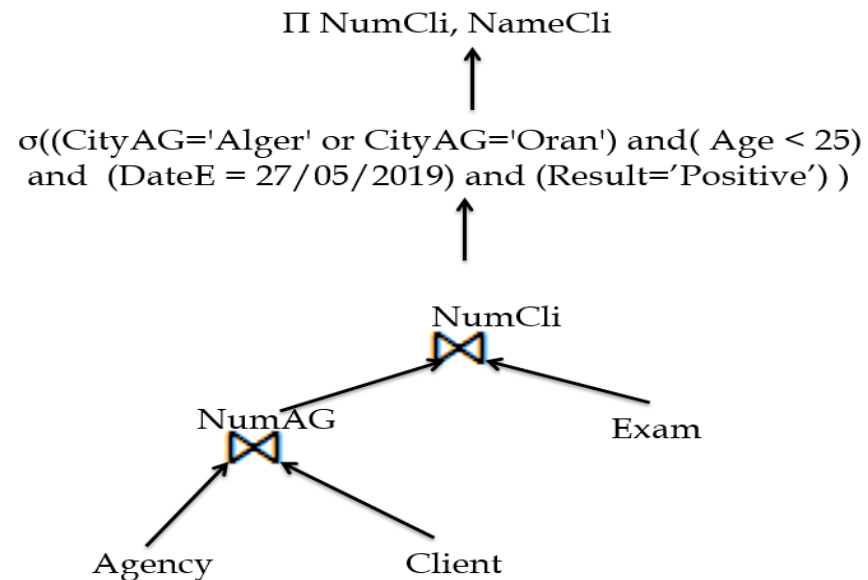
1- SQL Query:

```
Select NumCli, NameCli
From AGENCY A, CLIENT C, EXAM E
Where (A.CityAG='Algeirs' or A.CityAG='Oran')
and (C.Age < 25)
```

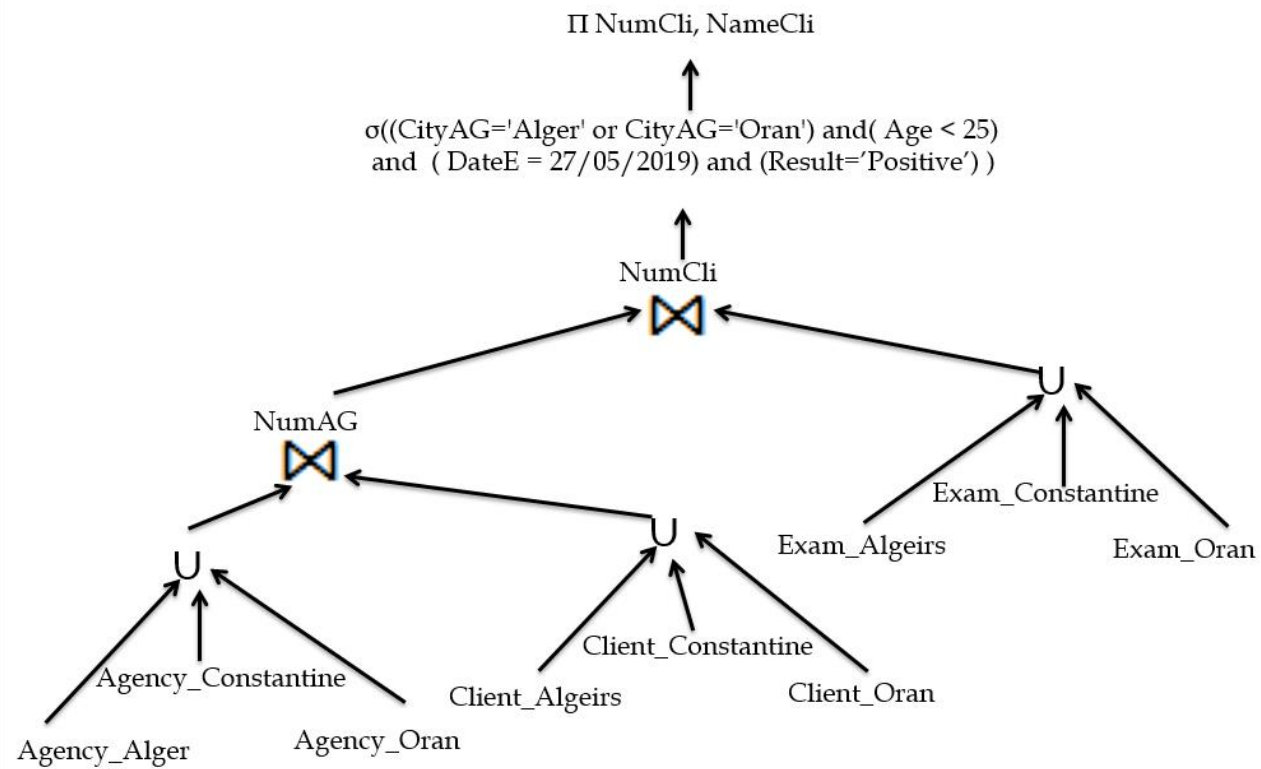
and (E. DateE = 27/05/2019)
and (E.Result='Positive')
and (C.NumAG = A.NumAG)
and (C.NumCli = E.NumCli);

2- Algebraic global tree:

➤ Global Tree:

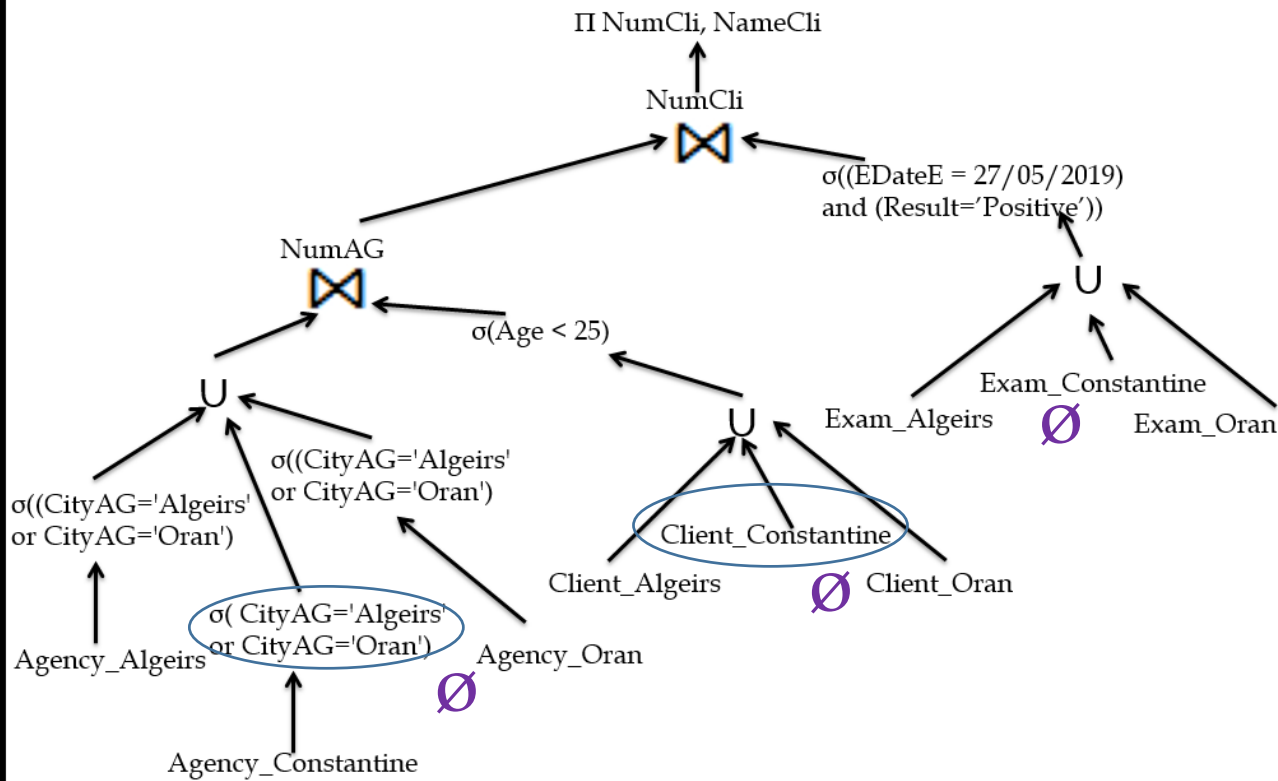


3- Canonical Tree

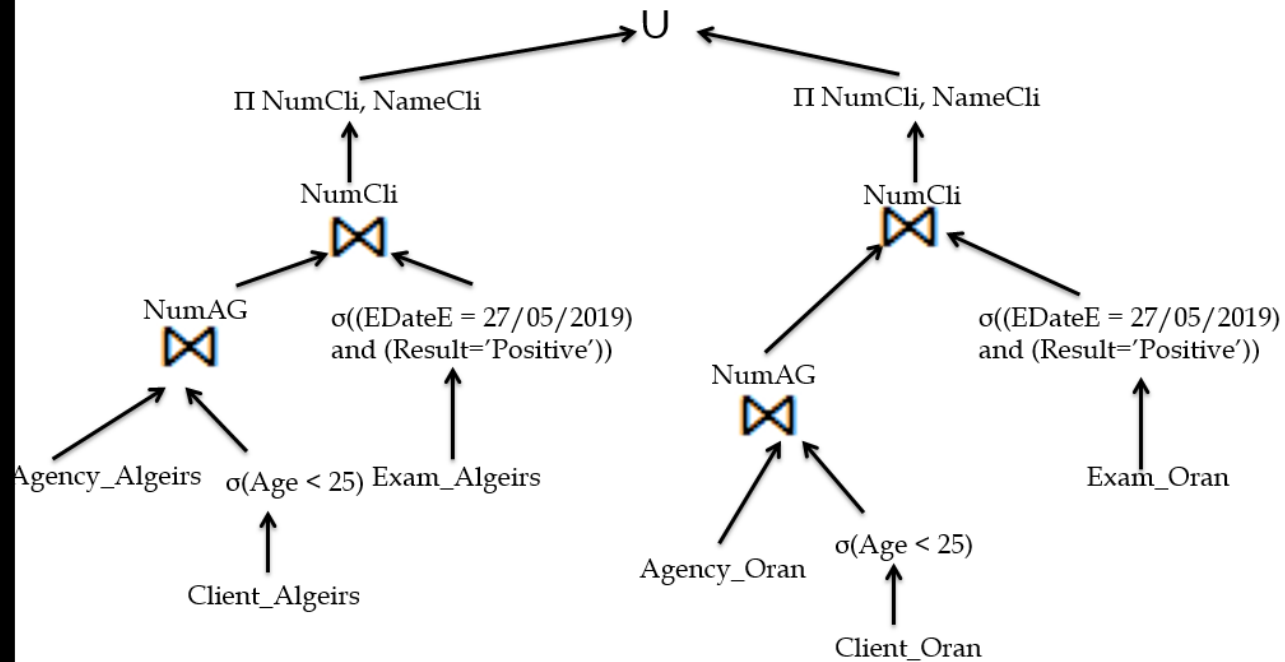


4- Optimized tree with general optimisation rules

➤ General optimisation rules:



- Distribution of the join operator according to the union and elimination of the empty fragments:



5- b. Give two execution strategies for the query Q

Strategy 1:

Execute the left part of the previous tree in Algiers

Execute the right part in Oran and send the result to Algiers

Join the two results

Strategy 2:

Send the fragments Agency_Oran, Client_Oran, and Exam_Oran to Algiers

Execute the entire query in Algiers